

TAIZHEN CHEUNG

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PROFESSIONAL SUMMARY

Software engineer and Computer Science + Mathematics student (May 2027) building reliable AI evaluation, retrieval, and full-stack systems. Experienced with Python, TypeScript, React/Next.js, FastAPI, PostgreSQL, LLM benchmarking, RAG, and automated testing.

EDUCATION

Hunter College, CUNY | B.A., Double Major: Computer Science & Mathematics

New York, NY | Expected May 2027

Relevant Coursework: Data Structures, Algorithms, Software Analysis & Design, Numerical Methods, Linear Algebra, Multivariable Calculus, Discrete Mathematics

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Frameworks & AI: React, Next.js, Preact, Node.js, Express.js, FastAPI, Pydantic, PyTorch, Scikit-learn, NumPy, Pandas

Data & Tools: PostgreSQL, Supabase, pgvector, SQLite, Docker, Git, GitHub Actions, Linux/Unix, pytest, Ruff, Playwright, REST APIs, MCP

Core Concepts: Data Structures, Algorithms, OOP, Machine Learning, LLM Evaluation, RAG, Vector Search, Reciprocal Rank Fusion, System Design, CI/CD

EXPERIENCE

Handshake | AI Trainer - Software Engineering & Coding Domain

Remote | Jun 2026-Aug 2026

- Evaluated frontier LLM responses across multi-turn software engineering tasks, scoring correctness, instruction following, technical communication, and helpfulness against detailed rubrics.
- Authored evidence-based rationales and comparative preference evaluations; calibrated judgments to support high inter-rater agreement and human-feedback data quality.

Handshake | AI Trainer - Image Processing Domain

Remote | Nov 2025-May 2026

- Used Python tooling, structured annotation, and rubric-based QA to validate multimodal datasets, document inconsistencies, and surface bias risks in image-processing and computer-vision examples.

TIER Lab, Hunter College | Research Intern - LLM Bias & Benchmarking

New York, NY | Jun 2025-Sep 2025

- Benchmarked 1,000+ prompts across Gemini 2.0 Flash, DeepSeek R1, and GPT-4o for reasoning consistency, safety, latency, and bias using reproducible Python datasets.
- Designed bias-audit protocols for LLM-integrated robotics and human-robot interaction, quantified latency/bias tradeoffs, and delivered mitigation recommendations.

Capital One | First-Gen Focus Fellow

Oct 2024-Dec 2024

- Selected for a competitive eight-week technical program covering system design, Agile development, and the software development lifecycle in collaboration with Capital One engineers.

SELECTED PROJECTS

Onward | Next.js 15, TypeScript, Supabase, PostgreSQL, LLMs

May 2026-Present

- Built a production full-stack storytelling application with SSE streaming, a client state machine, provider-pluggable LLM/embedding interfaces, Supabase Auth, session ownership, and atomic per-user/IP rate limiting.
- Developed a six-lane vector/RRF/LLM-reranked retrieval challenger, then retained the stronger production keyword+rereanker recipe after it achieved 99/101 top-1 results with zero definitive-wrong outcomes.

When Directional Accuracy Lies - Sole-Author Preprint | [arXiv:2607.12248v2](https://arxiv.org/abs/2607.12248v2)

2026

- Authored and posted a 10-page arXiv preprint presenting a base-rate-honest benchmark for LoRA-adapted TimesFM equity forecasting with leakage-controlled expanding walk-forward evaluation.
- Evaluated frozen S&P 500 and NASDAQ-100 universes; released checksum-verified data, reproducibility tooling, and 55 CPU tests, finding no directional skill over the always-up base rate.

resufill | Python, FastAPI, Pydantic, Preact, TypeScript, Playwright

Aug 2026

- Built a local CLI and web UI that transforms a canonical career record into job-specific ATS-parseable resumes and cover letters while tracing every bullet to a source ID.
- Implemented unsupported-claim rejection, missing-keyword gap reporting, claim-by-claim audits, and Chromium PDF render/parse-back verification; maintained 207 automated tests.

xiaofinance | Python, FastAPI, SQLite, Preact, TypeScript, MCP

Jul 2026

- Built a local pipeline and dashboard that analyzes the latest 24 hours of Xiaohongshu investment discussion, with simhash repost deduplication and conversation-volume scoring.
- Implemented context-gated Chinese alias and retail-slang detection, verbatim quote validation for LLM summaries, and an MCP server for grounded agent analysis.